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BACKGROUND

- Globally, the impact of climate change is increasingly evident, manifesting in more frequent and intense extreme weather events, including floods(UNDRR, 2021). These events pose significant threats to human lives, infrastructure, and economic stability, demanding effective mitigation strategies.
- Regionally, East Africa is particularly vulnerable to the effects of climate change. Heavy rains and subsequent flooding have become recurring challenges, impacting communities and hindering development across the region (Openjuru & Ph, 2019).
- Within this context, River Nabuyonga, located in Eastern Uganda, presents a stark example of the devastating consequences of flooding. The river, originating from the slopes of Mount Elgon and flowing through various regions including Mbale City, is infamous for its history of destructive floods, earning it the somber nickname "the river of death"(For & City, 2022). These floods are primarily caused by heavy rainfall, environmental degradation, rapid urbanization, and climate change, which increase runoff and overwhelm the river's capacity consequently leading to flash floods along the riverbank. The effects on the community include loss of life and property, displacement, economic losses, and health risks from waterborne diseases. The 2022 flood incident along River Nabuyonga in Mbale District was particularly devastating.

Cont.

- Triggered by heavy rains on the slopes of Mount Elgon, the floods occurred on the night of July 30th and continued into the morning of July 31st. The intense rainfall caused the river to overflow, leading to widespread destruction. At least 24 people lost their lives, with many others reported missing. The floods submerged homes, shops, and roads, causing significant damage to infrastructure and property (UNDRR, 2021). These recent events, highlight the urgent need for comprehensive flood control measures along the Nabuyonga River. Furthermore, a recent flash flood event in Mbale city, triggered by heavy rainfall on Thursday, November 28th 2024, highlights the urgency of effective flood mitigation measures. The flash floods caused widespread disruption, rendering roads impassable and severely impacting the daily lives of residents

- To control the effect, measures such as improved urban planning, reforestation, early warning systems, infrastructure development, and community education are being implemented (Rosa Malango, 2020). These efforts aim to enhance resilience and reduce the impact of floods, though continuous improvement and adaptation are necessary to effectively mitigate the risks. However, there is a significant research gap on the analysis of structural and non structural flood control measures to further combat the increasing fooding extent.

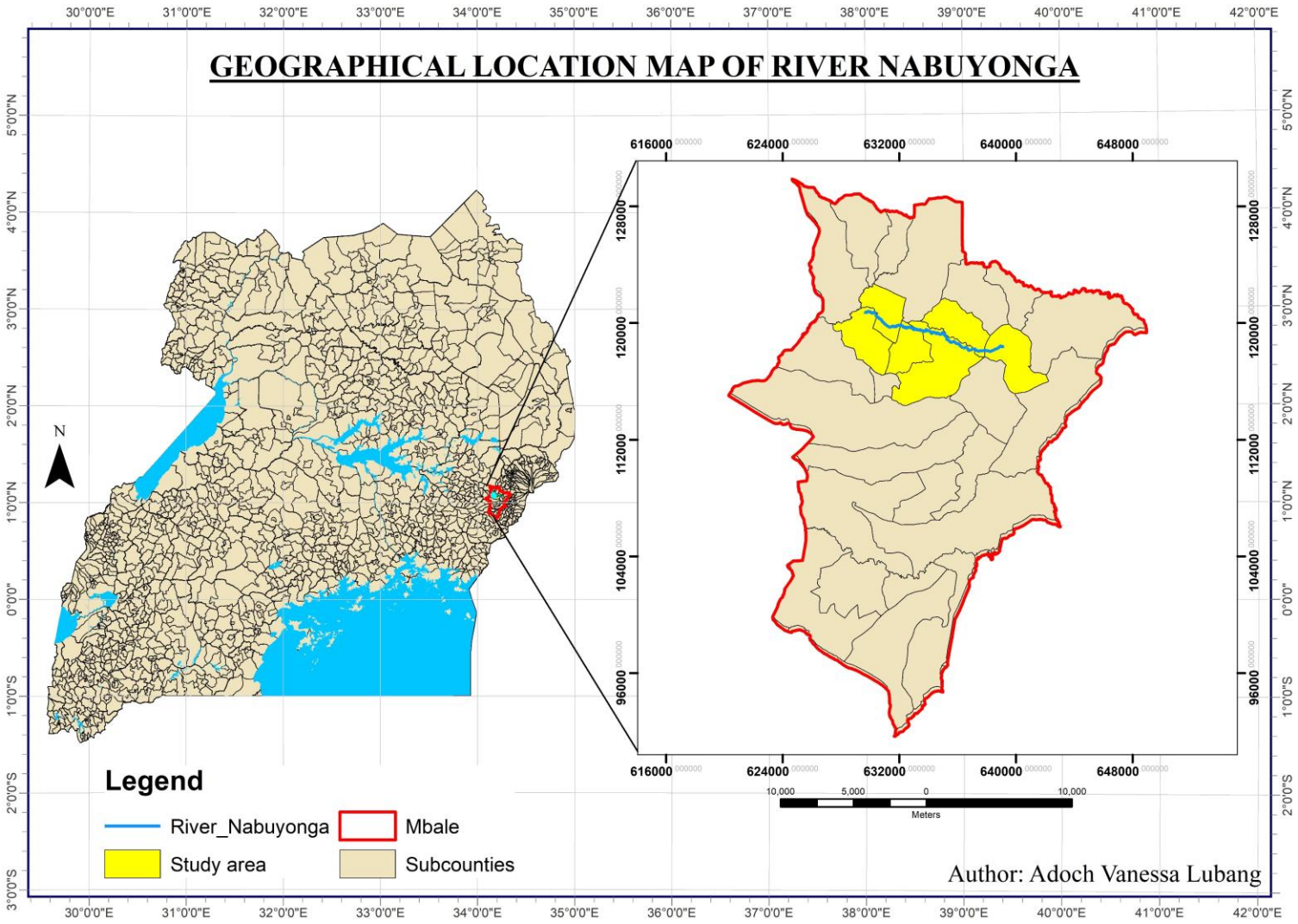
PROBLEM STATEMENT

- Despite the implementation of the Bamboo planting initiative, there is a notable knowledge gap in assessing the efficacy of commonly implemented flood control measures in reducing spatial flooding extent on River Nabuyonga (Wali et al., 2013). This knowledge gap is attributed to the limited consideration of the river hydrology, land use and land cover, and topography of the study area in the design and implementation of flood control measures for the context of the Nabuyonga catchment and its unique characteristics. (Foufoula-Georgiou et al., 2014; Sene, 2015). Furthermore, current research on flood control measures especially levees and riparian buffers often overlooks the long-term performance and flooding extent reduction of these measures in regional communities such as Mbale city , leaving a significant gap in our understanding of their true efficacy (Kirono et al., 2010).

- This study aims to bridge this **knowledge gap by conducting a comprehensive evaluation of the effectiveness of levees and riparian buffers in reducing spatial flooding extent on River Nabuyonga**. A mixed-methods approach will be employed, integrating hydrological modeling (HEC-RAS), Buffer analysis using SWAT modelling, geospatial analysis (ArcGIS), and as well as LSTM (Long short term memory) time series analysis because of its high level of accuracy and reliability. This integrative approach will enable the quantification of flood extent reduction, identification of key factors influencing flooding, and assessment of the long-term performance of flood control measures.

STUDY AREA

- The study area is the River Nabuyonga catchment, with a focus on the 200m floodplain buffer along the river.
- This area is selected due to its history of flooding and the need for effective flood mitigating measures.
- The catchment's unique characteristics such as its topography, land use and soil type it ideal for studying the impact of riparian buffers and levees on reducing the spatial flooding extent of the river.



OBJECTIVES

Main objective: To evaluate the effectiveness of levees and riparian buffers in reducing the spatial flooding extent on River Nabuyonga in Mbale city.

Specific objectives

1. To identify the factors that influence flood occurrences along the river Nabuyonga catchment.
2. To quantify the reduction in the floodplain achieved by levees and riparian buffers.
3. To determine the long-term performance of levees and riparian buffers on flood extent reduction.

Research questions

1. What Are The Primary Factors That Contribute To Flood Occurrences In The River Nabuyonga Catchment?
2. How Effective Are Levees And Riparian Buffers In Reducing The Flooding Extent On River Nabuyonga?
3. How will these measures perform over a 10-year period?

JUSTIFICATION

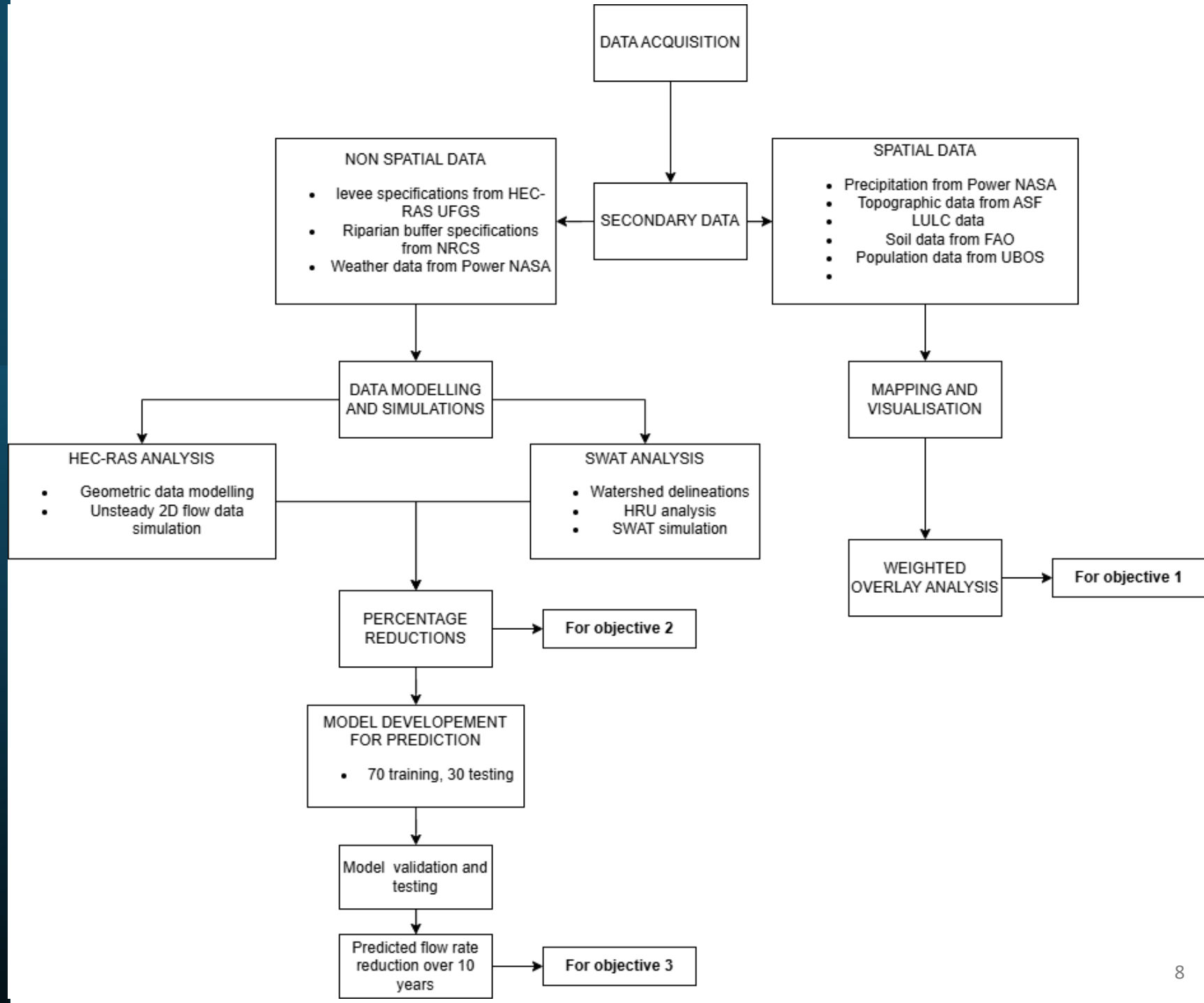
This research is significant to the community as it directly impacts the safety and economic stability of the neighboring communities. Improved flood control measures can prevent loss of life, reduce property damage, and minimize disruptions to local livelihoods. Additionally, understanding the environmental consequences of these structures can help maintain the ecological balance of the river and its surroundings, promoting sustainable development.

Moreover, this research aligns with several Sustainable Development Goals (SDGs), including:

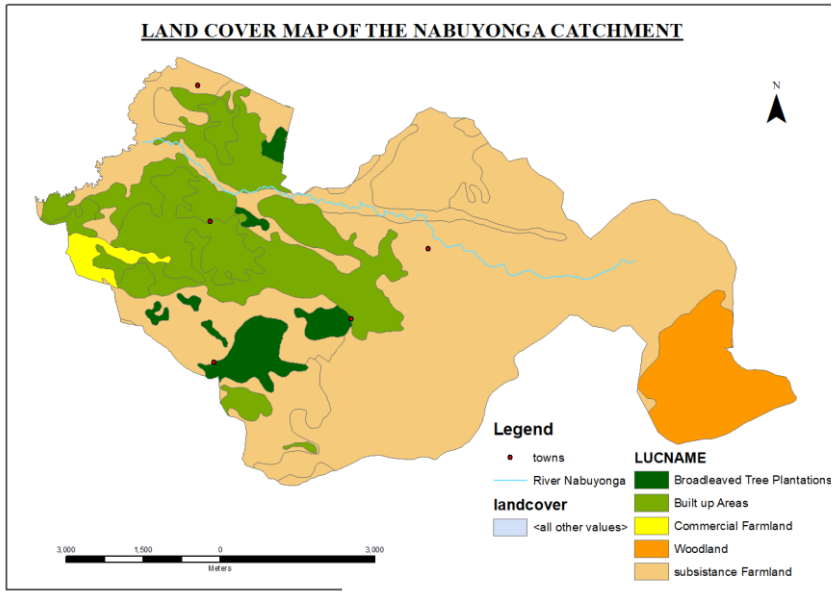
- SDG 6 (Clean Water and Sanitation) which ensures sustainable management of water resources.
- SDG 11 (Sustainable Cities and Communities) which is responsible for making cities and human settlements inclusive, safe, resilient, and sustainable.
- SDG 13 (Climate Action) which is accountable for strengthening resilience and adaptive capacity to climate-related hazards.

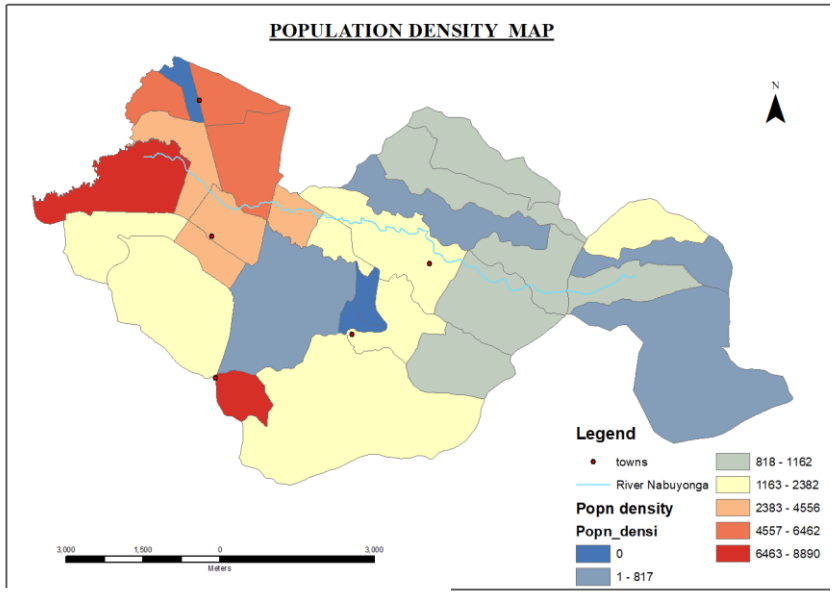
By addressing these goals, the research not only contributes to local community resilience but also supports global efforts towards sustainable development.

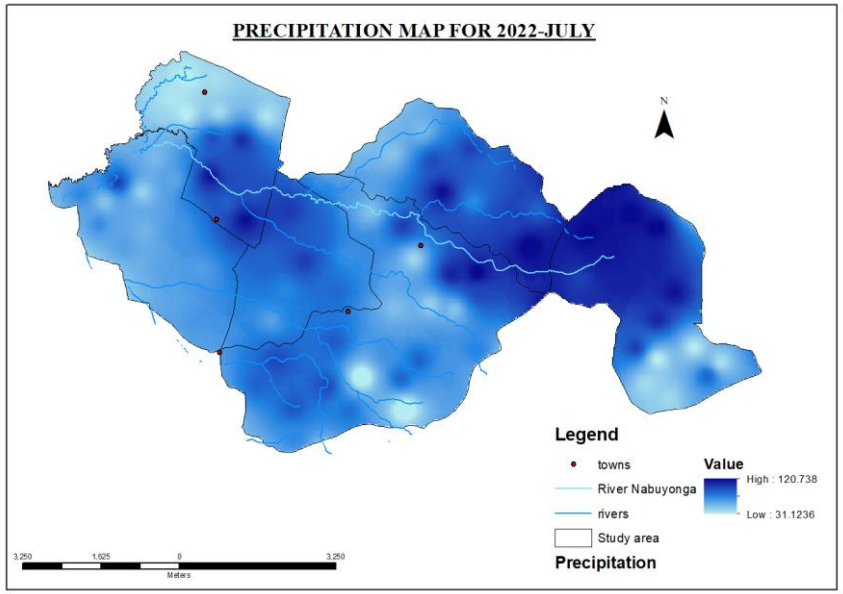
METHODOLOGY

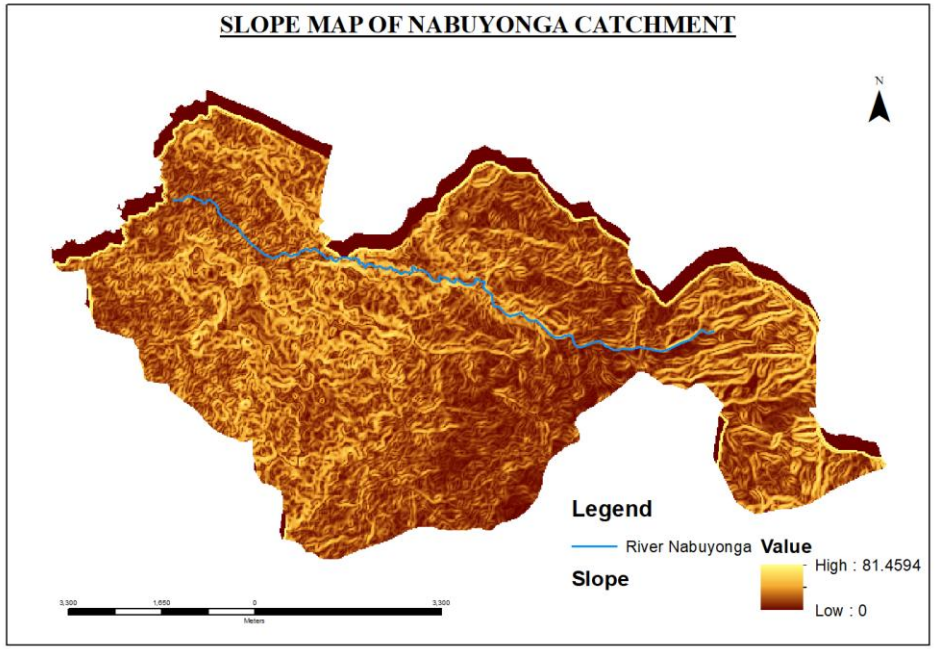


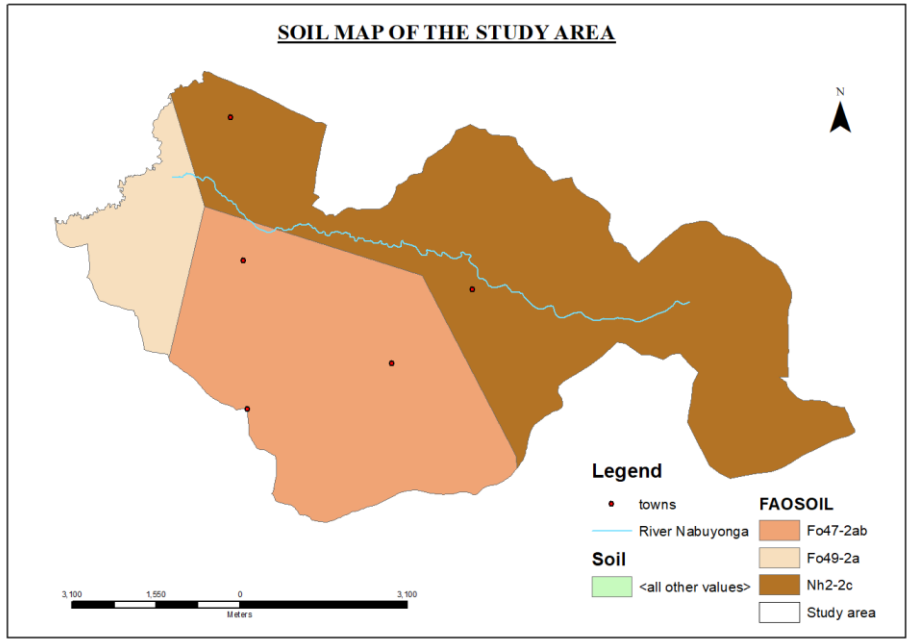
OBJECTIVE ONE : To identify the factors that influence flood occurrences along the river Nabuyonga catchment



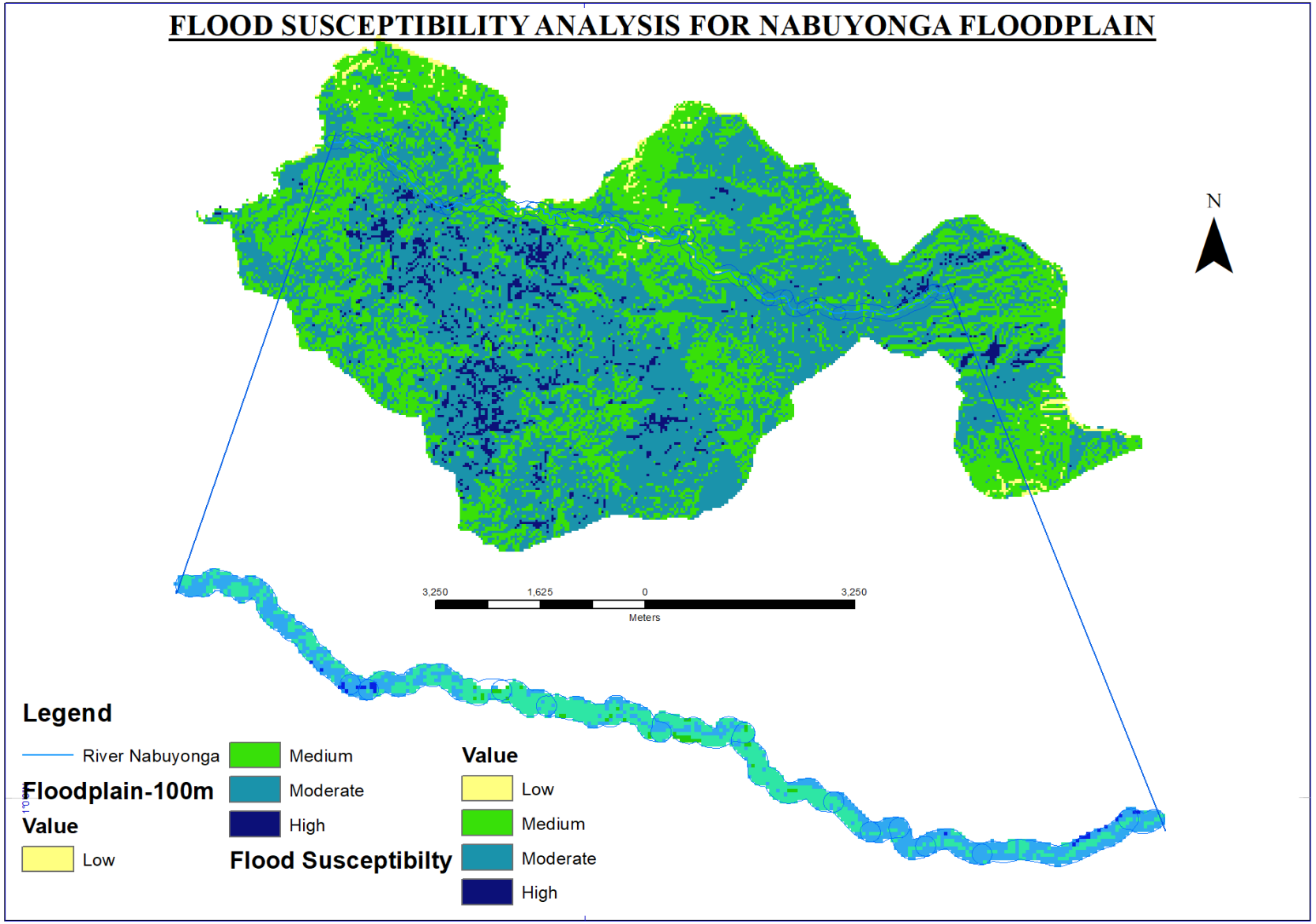


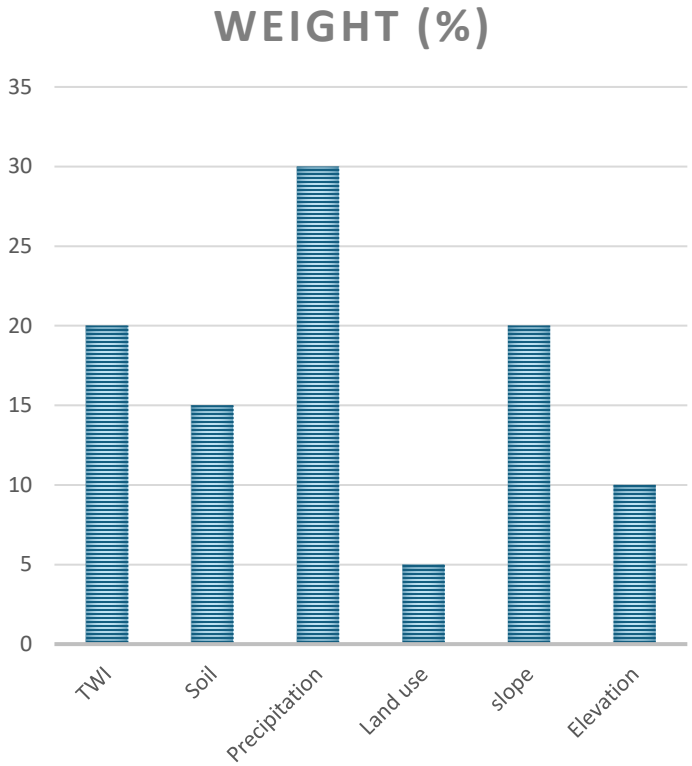






WEIGHTED OVERLAY ANALYSIS





OBJECTIVE 1:Results and conclusion

From the analysis,

- ***Precipitation(30%):*** Heavy rainfall during the flash floods overwhelms the drainage capacity and bursts the riverbanks. The Nabuyonga catchment locations and climate make the river prone to intense rainfall events which contributes to the high percentage.
- ***TWI (20%):*** The Nabuyonga catchment topography, with its low-lying areas and convergent flow paths, contributes to the significant influence of the TWI on flood occurrences.
- ***Slope (20%):*** Nabuyonga catchment has varied slope gradients that contribute fooding events due its gentle slopes which make it more prone to stagnations and flooding.
- ***Soil (15%)*** : Soil characteristics of soils predominant in the Nabuyonga catchment such as sandy clay soils which have a low infiltration capacity and high impermeability which lead to flooding event during intense rainfall events.
- ***Elevation (10%):*** Elevation affects flow direction and accumulation of water. Since the Nabuyonga catchment has a low elevation therefore water tends to flow to these areas.
- ***Land use (5%):*** Predominant land use patterns such as built-up areas and subsistence farming alter the hydrological response of the watershed and distract the channels for surface runoff and redirects them to flooding basins along the river.

Therefore,

The weighted overlay analysis successfully identified the factors that influence flooding occurrences and provided valuable insights for flood risk management.

OBJECTIVE TWO: To quantify the reduction in the floodplain achieved by levees and riparian buffers.

HEC-RAS ANALYSIS

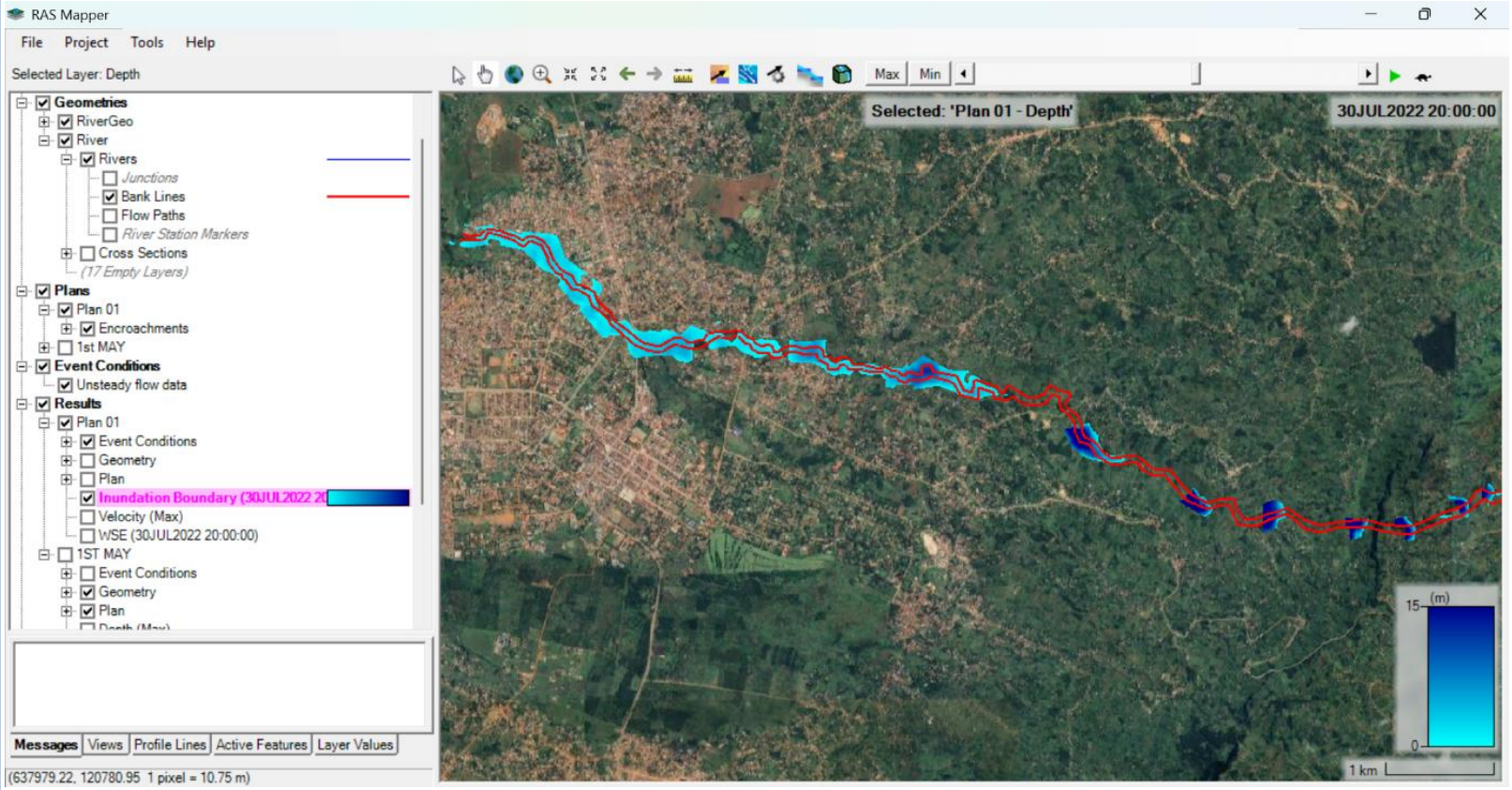
This objective will be achieved with hydraulic modeling and spatial analysis techniques to quantify the reduction in spatial flooding extent with levees and riparian buffers implementation.

Data requirements

- DEM
- Land use (ManningsN value)
- Google satellite image
- River Nabuyonga Flow rate data

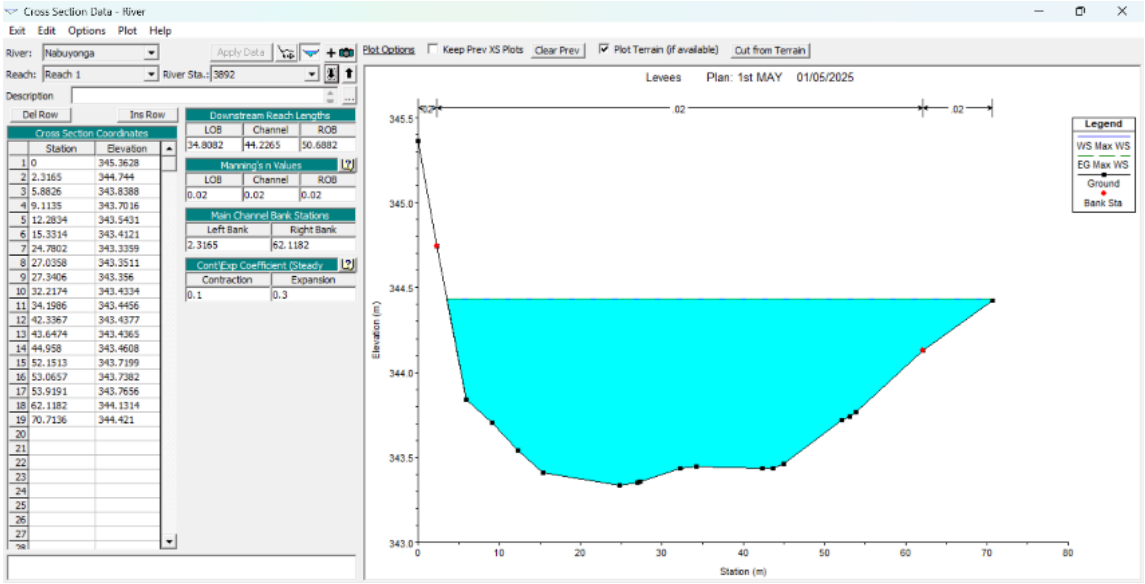
Analysis methods

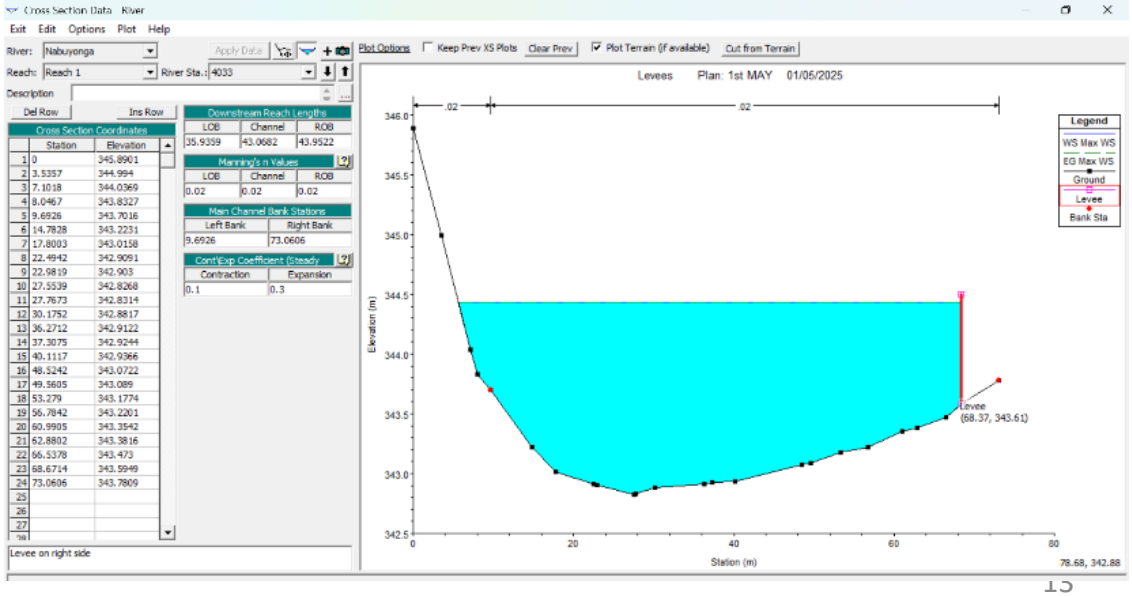
- Flood event simulation (Geometric data: cross sections, flow path, bank lines)
- Unsteady flow analysis (2022 flow rate data, ManningsN values)
- Levee installation



Levee Installation

River station	Left Bankline	Right Bankline	Levee location			Spatial reduction(m)
			Station	Elevation	Bankline location	
2907	62.484	56.6928	10.66	342.46	ROB	10.66
3737	83.0275	50.5663	34.57	345.42	LOB	32.45
3892	34.8082	50.6882	61.89	345.12	ROB	8.78
3208	37.6428	50.3225	4.80	342.16	ROB	5.03
4033	35.9395	43.9522	68.59	343.60	LOB	4.83
4632	48.2194	46.1772	6.98	345.35	LOB	6.98
4910	34.991	49.0423	18.22	346.43	LOB	18.22
5021	35.5702	28.7122	5.27	346.77	LOB	5.27
			67.92	346.75	ROB	7.1
5671	46.5734	49.8043	55.09	349.38	LOB	18.65
5859	56.9976	48.4937	71.19	350.24	LOB	13.31
9521	73.0606	38.6182	70.07	410.98	LOB	13.47
9719	70.0735	51.4502	59.95	415.33	LOB	23.58
9980	55.687	97.2312	16.37	426.76	ROB	16.37
10390	80.3148	60.4418	12.95	453.45	ROB	12.95
10794	53.7972	65.1662	22.65	526.05	ROB	22.65
11767	41.8186	63.7032	14.57	573.33	ROB	14.57

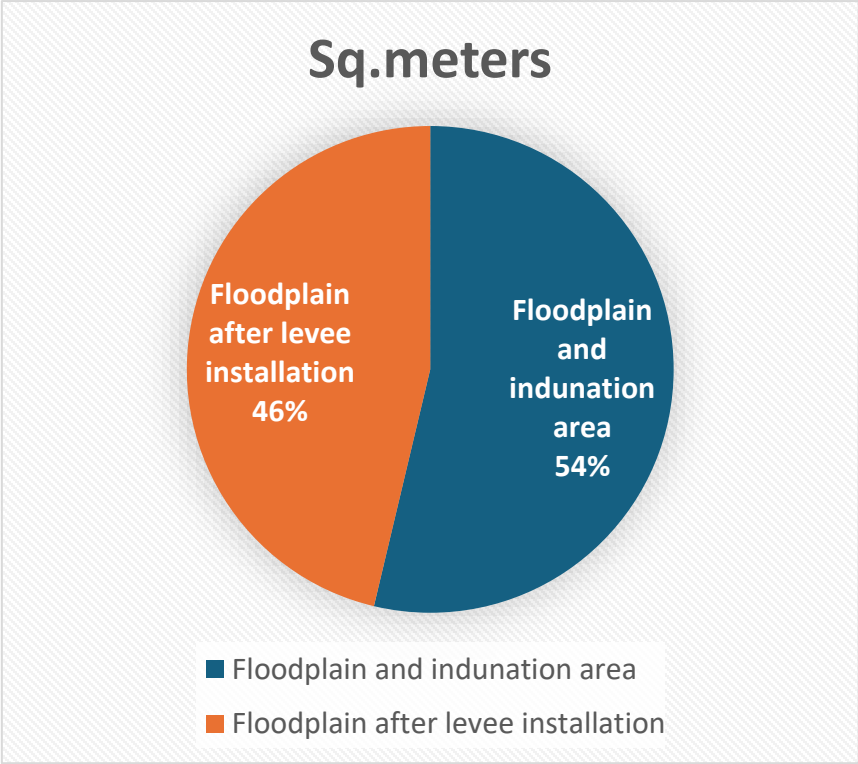


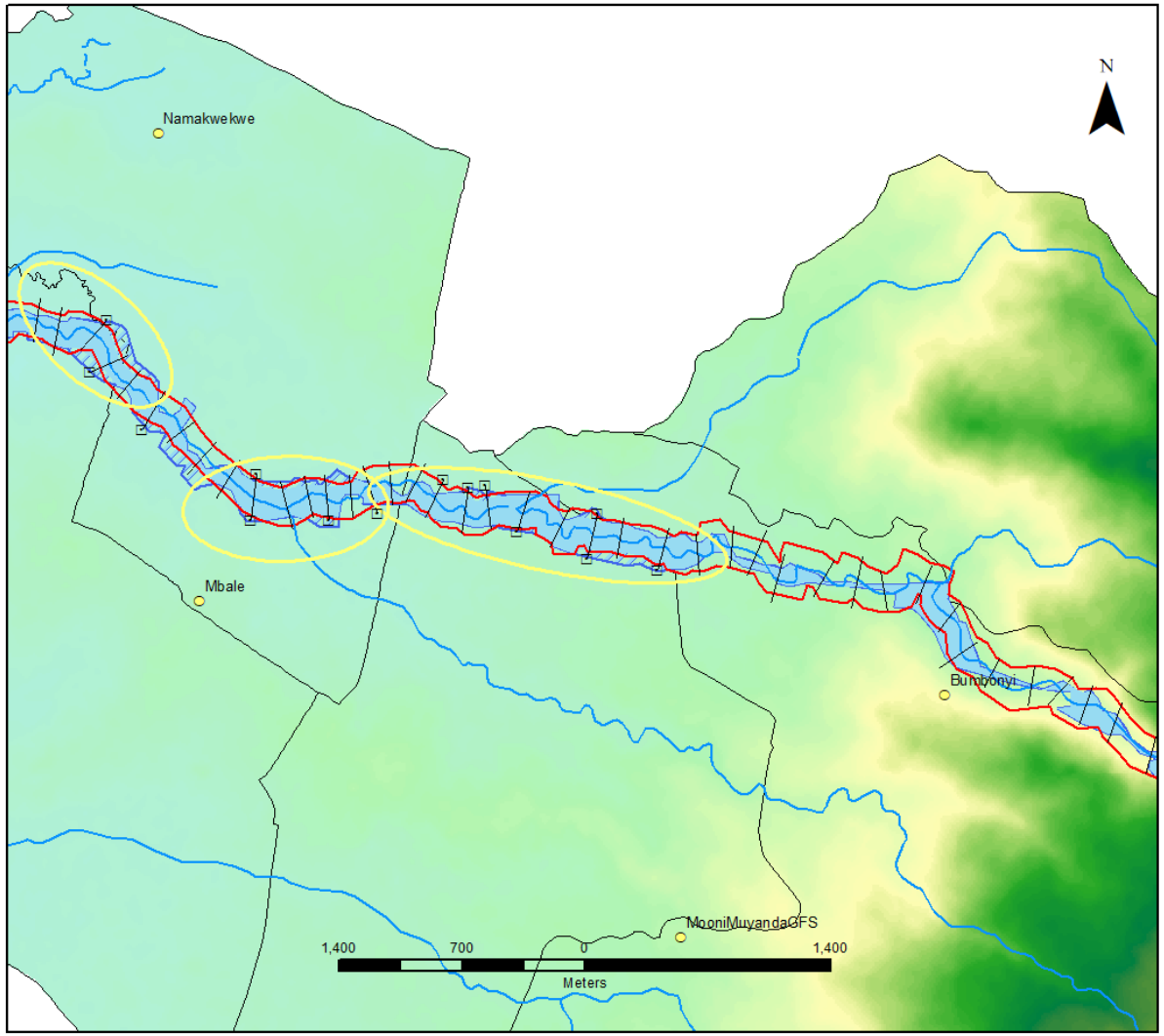


Levee installation effect on floodplain

Levee installation parameters

- Pre-installed levee specifications in HEC-RAS software
- Levees were installed at 16 stations on the right or left banks.
- The average spatial flood reduction is 14.67 m.





LEVEE INSTALLATION ON FLOODPLAIN

Legend

- Levees
- Flood region
- rivers
- Floodplain
- With levees
- River inundation
- Study area

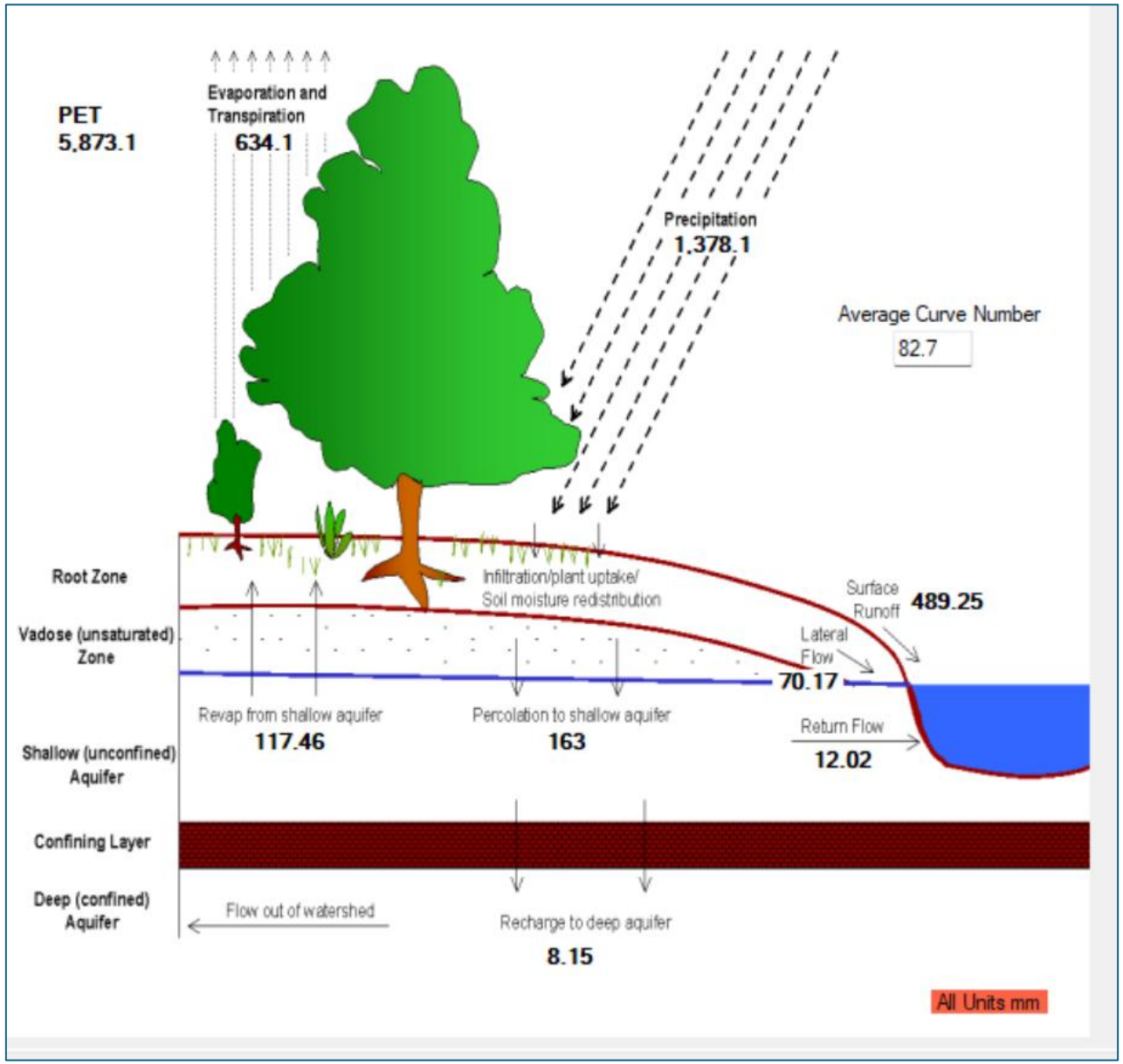
dem

Value

High : 2194

Low : 1080

Part 2: SWAT analysis and simulation



For the SWAT analysis

Data requirements

- LULC
- Soil
- Slope
- Weather parameters (temperature, solar radiation, rainfall, wind, relative humidity,)

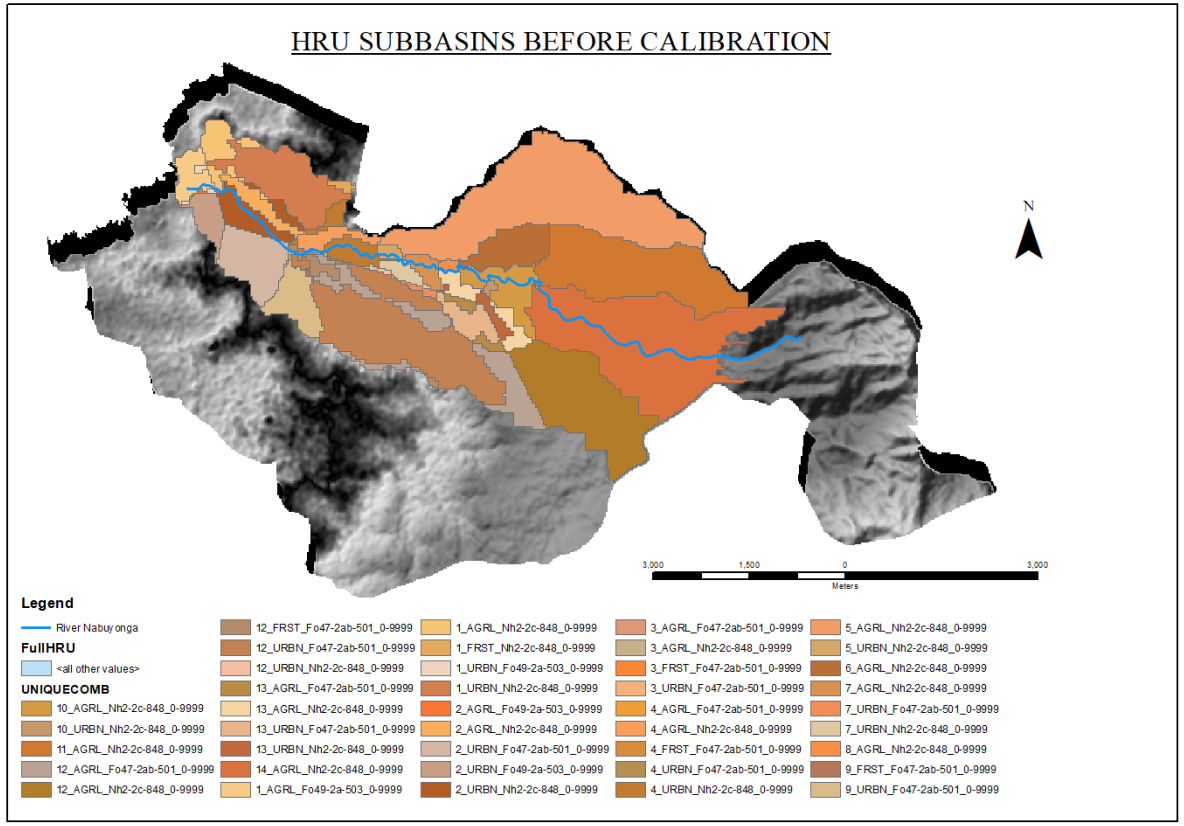
Data analysis

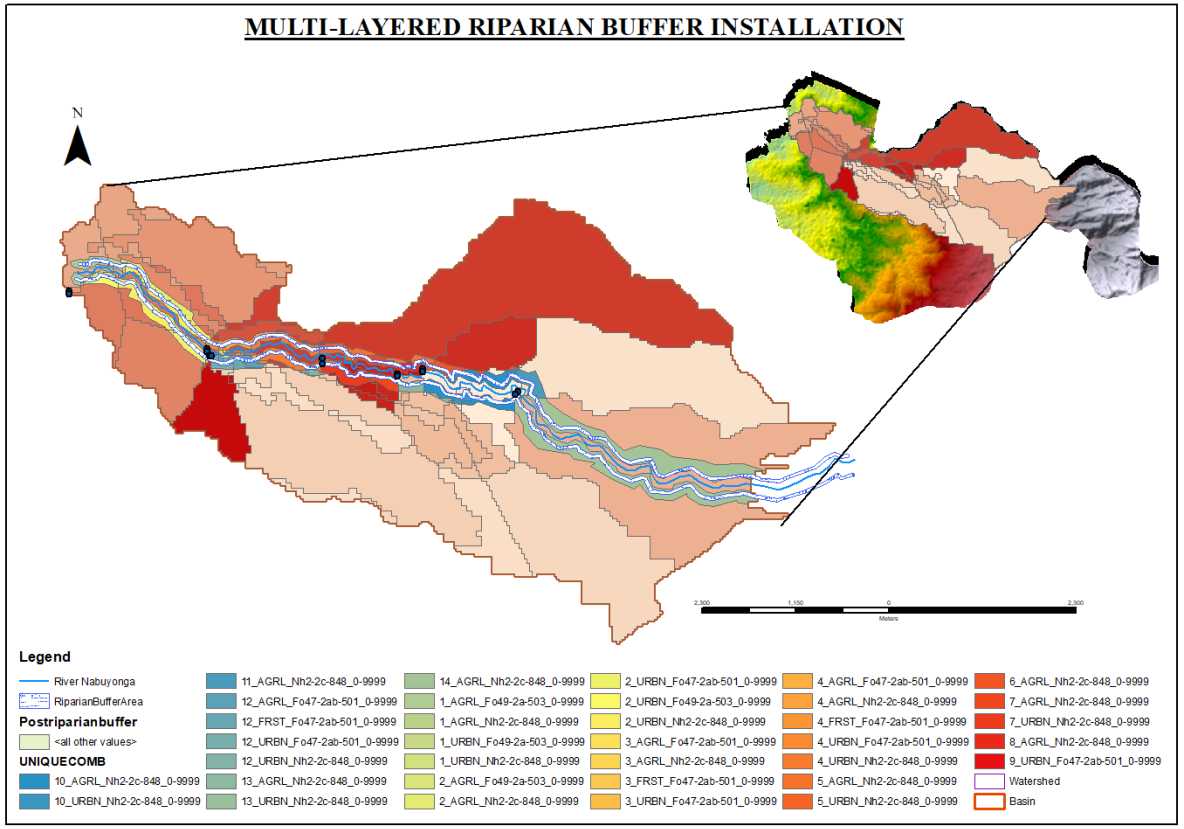
- Watershed delineation
- HRU analysis
- Write input tables
- SWAT simulation

Watershed simulation before calibration

Simulation length	**13 yrs**
Warmup	1 yr
HRUs	30
Subbasins	14
Output timestep	Daily
Precip. method	Measured
Watershed Area	26.01 sq.km

Nabuyonga catchment before and after calibration





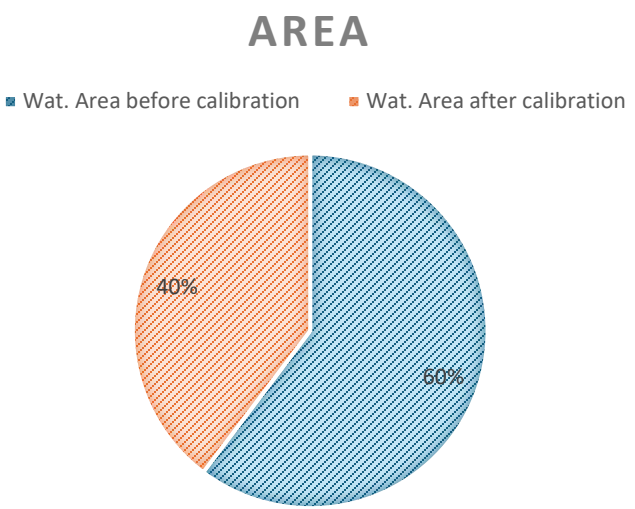
Riparian buffer parameters

- 20m floodplain buffer for riparian buffer boundary
- Multi-layered riparian buffer of mixed vegetation of grasses and trees.

Vegetation	Sediment Filtration rate (%)	Runoff-reduction rate (%)
Dogwood	60	35-90
Willow	44	45-76
Switch grass	70	20-78
Bamboo	80	15-65
Fescue grass	93	11-79

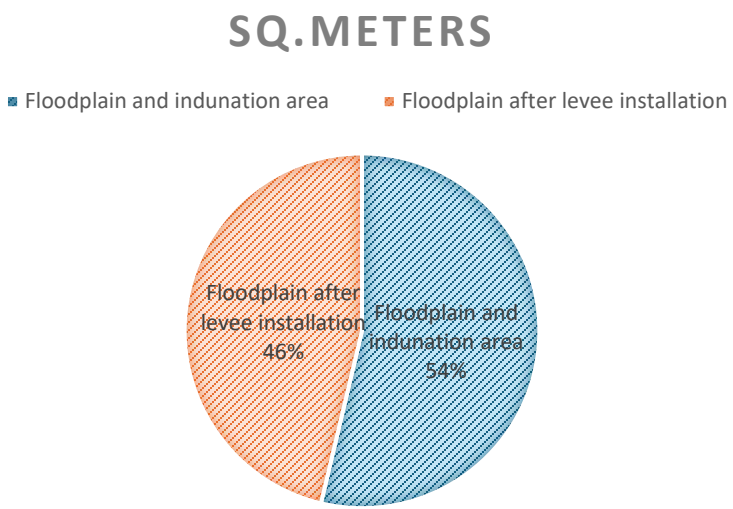
Objective 2: Results and conclusion

Riparian buffer installation



- 95% Confidence interval - 18.04%, 21.96%
- MSE - 150

Levee installation

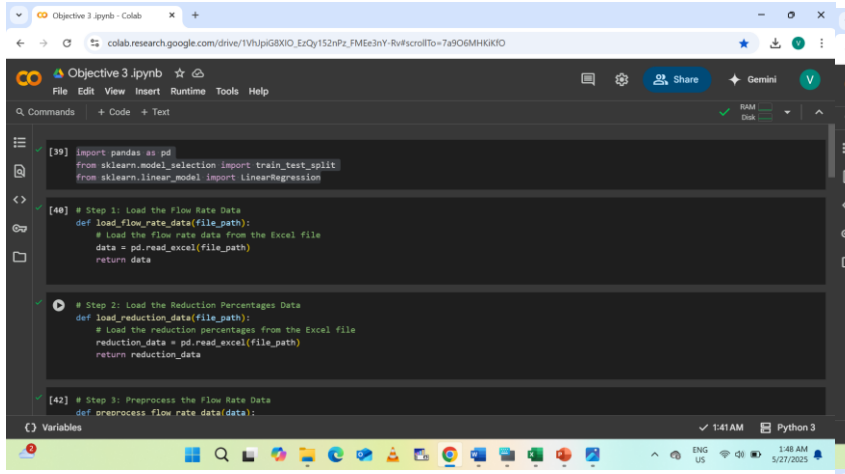


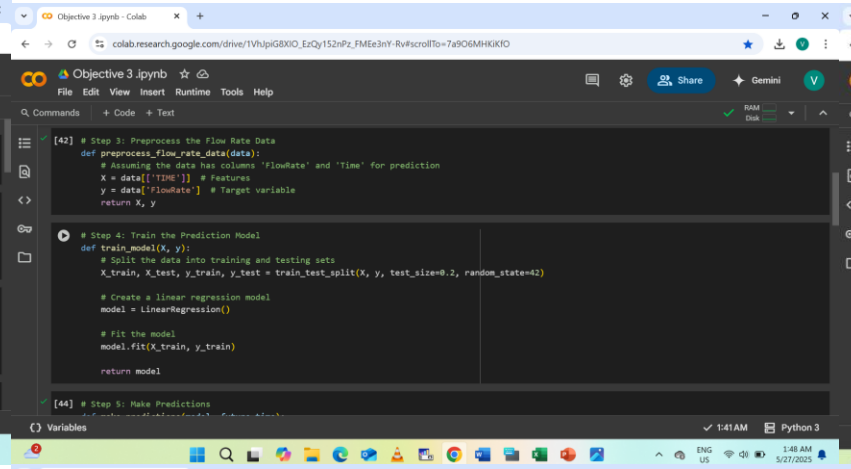
- Confidence interval – 11.04%, 14.96%
- MSE – 200

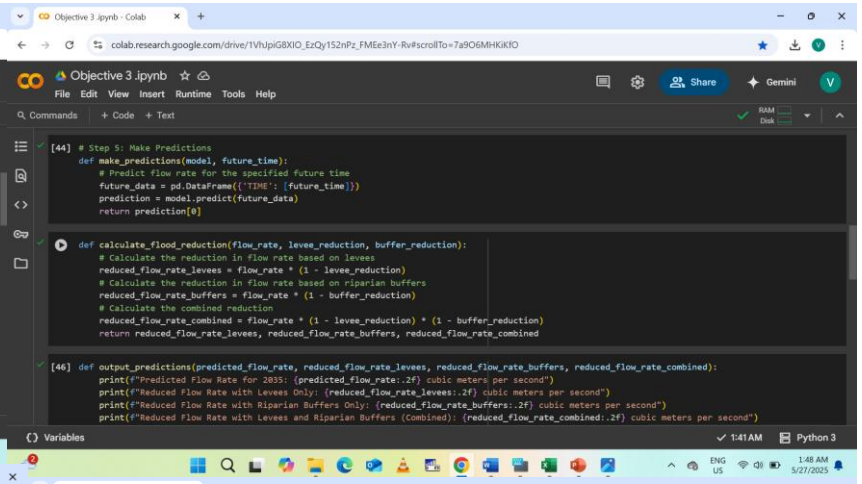
- The riparian buffers reduce the spatial flooding extent by 20.348% and levees by 13.848%.
- The variations in these results show how effective riparian buffers work as a flood control measure as opposed to levees.
- Riparian buffers extend the river floodplain and act as an ecological barrier between the other land use classes and the riverbanks.
- Levees on the other hand create a hard barrier between the riverbanks and floodplain which increases the river flow rate as realized in Objective 3.

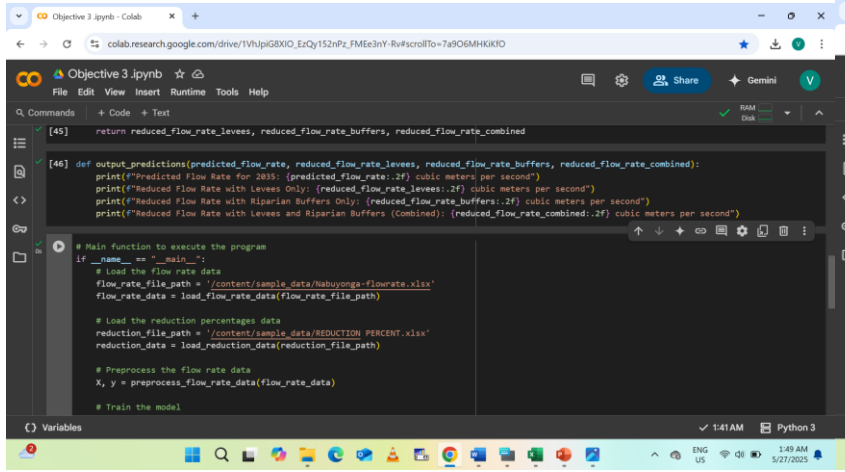
OBJECTIVE 3: To determine the long-term performance of levees and riparian buffers on the flood flow rate

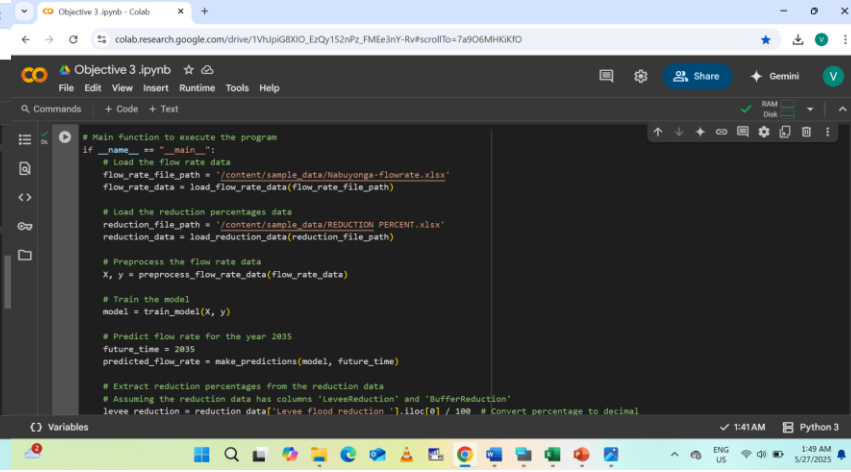
- For this objective, datasets for the flow rate and percentage reductions induced with levee and riparian buffers installation were fed into the model in Google colab.

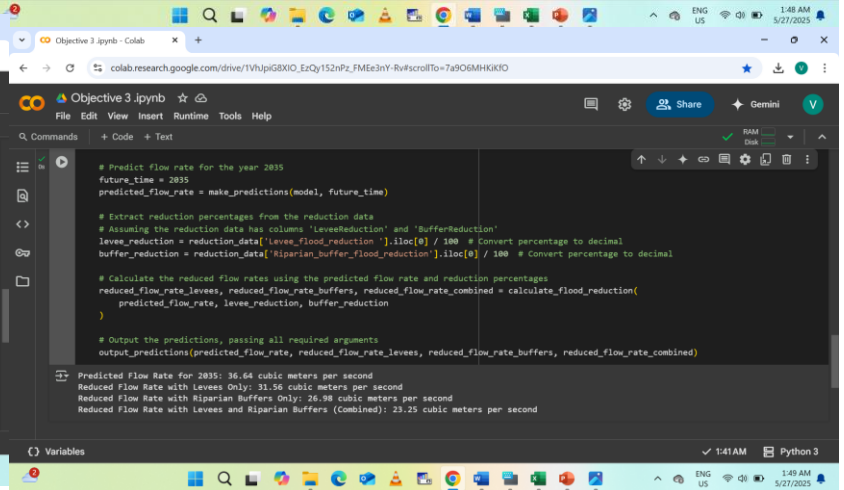






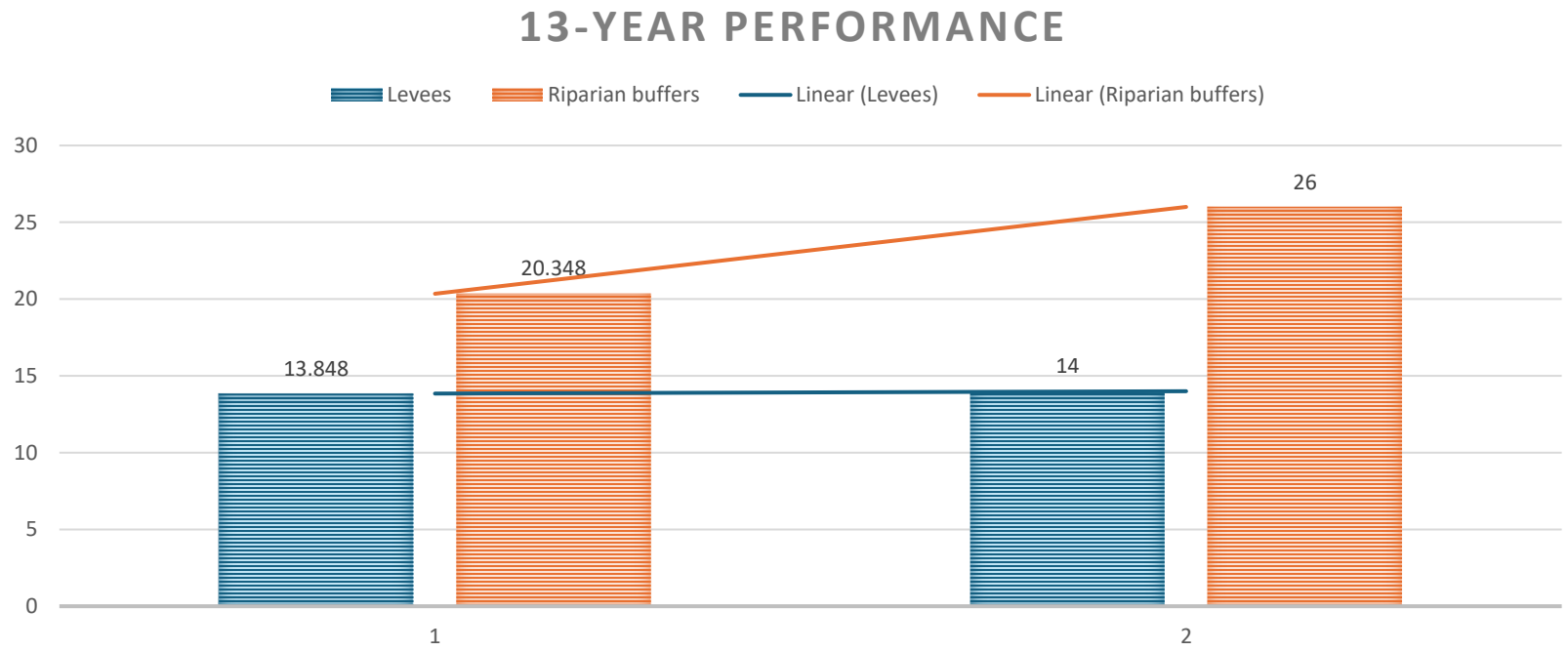






Objective 3: Analysis of results

- Predicted Flow Rate for 2035: 36.64 cubic meters per second
- Reduced Flow Rate with Levees Only: 31.56 cubic meters per second
- Reduced Flow Rate with Riparian Buffers Only: 26.98 cubic meters per second



Key findings

- The predicted flowrate for 2035. The model predicts a flow rate of 36.64 m3/sec which indicates a potential flood risk. However, a negative correlation is noticed due to implementation of both riparian buffers and levees.
- Effectiveness of levees. The implementation of levees is expected to reduce the flow rate to 31.56 m3/sec, representing a reduction of 14 % compared to the predicted flow rate in 2035.
- Effectiveness of riparian buffers. The use of riparian buffers is predicted to reduce the flow rate to 26.98 m3/sec, representing a reduction of approximately 26% as compared to the predicted flow rate.

RESEARCH CONCLUSIONS

Key Factors Influencing Flooding (Weighted Overlay Analysis)

- **Precipitation (30%)** – Most significant factor
- **TWI (20%)** & **Slope (20%)** – Moderate influence
- **Soil (15%)**, **Elevation (10%)**, **Land Use (5%)** – Lesser impact

Floodplain Reduction Analysis

- **Levee installation:**
  - Reduced floodplain area from **54% to 46%**
  - Implemented at **16 stations**
  - **Average spatial flood reduction: 14.67 m**
- **Riparian buffers:**
  - Reduced water area from **60% to 40%**
  - **Overall spatial flood reduction: 20.348%**
  - More effective than levees in expanding floodplain & acting as an ecological barrier.

Long-Term Performance on Flood Flow Rate (Predictions for 2035)

- **Without intervention: Flow rate = 36.64 m^3/sec** (high flood risk)
- **Levees impact:**
  - Expected flow rate **reduction to 31.56 m^3/sec** (**14% decrease**)
  - Hard barrier approach increases river flow rate over time

Riparian buffers impact:

- Expected flow rate **reduction to 26.98 m^3/sec**
- More effective in controlling flood flow long-term

Overall Conclusion

- **Riparian buffers outperform levees** in reducing both **spatial flooding extent** and **long-term flood flow rate**
- Riparian buffers provide **ecological benefits** and allow natural floodplain expansion
- Levees create hard barriers but may **increase river flow rate over time**

RECOMMENDATIONS

- ***Prioritize Riparian Buffers*** – More effective than levees in reducing flooding and controlling long-term flow rates.
- ***Support Natural Solutions*** – Scale up the ongoing Bamboo planting initiative and engage communities and stakeholders in ecological flood control.

REFERENCES

Chereni, S., Sliuzas, R. V., Flacke, J., & Maarseveen, M. V. (2020). The influence of governance rearrangements on flood risk management in Kampala, Uganda. *Environmental Policy and Governance*, *30*(3), 151–163. https://doi.org/10.1002/eet.1881

For, D., & City, M. (2022). *2022 HEAVY DOWN POUR THAT RESULTED INTO A FLOOD*. *July*.

Gede Tunas, I., & Maadji, R. (2018). *The Use of GIS and Hydrodynamic Model for Performance Evaluation of Flood Control Structure*. *8*(6).

Genovese, E., & Thaler, T. (2020). *The benefits of flood mitigation strategies : effectiveness of integrated protection measures*. *6*(September), 459–472. https://doi.org/10.3934/geosci.2020025

Ghoraba, S. M. (2015). Hydrological modeling of the Simly Dam watershed (Pakistan) using GIS and SWAT model. *Alexandria Engineering Journal*, *54*(3), 583–594. https://doi.org/10.1016/j.aej.2015.05.018

Kitunzi, Y. (2024, December 1). Floods damaging Uganda's infrastructures, affecting lives, traffic to countryside. **Daily Monitor**. https://www.monitor.co.ug/uganda/news/national/floods-damaging-uganda-s-infrastructures-affecting-lives-traffic-to-countryside-4842070#story Mclean, L. (2015). *E-proceedings of the 36*. *1*, 1–9.

Mobley, W., Kwakkel, J. H., & Brody, S. D. (2019). *Evaluation of flood risk reduction strategies through combinations of interventions*. *12*(September 2018), 1–17. https://doi.org/10.1111/jfr3.12506

Muketha, S. M. (2019). *Riparian Zones and Their Role in Enhancing Resilience to Flooding in Urban Areas : A Case Study of Nairobi River Basin*. *13*(1).

Narura, A. K., Ssenku, J. E., Kudamba, A., Okurut, S. A., & Namuli, A. (2024). *The Environmental Conservation Strategies Used in Secondary Schools Align with Guidelines of United Nations Environmental Program : Evidence from a Cross-Sectional Study in a City Setting*. *23*(7), 200–230.

Openjuru, G. L., & Ph, D. (2019). *Makerere university makerere university*. *June*, 540009.

Remo, J. W. F., Carlson, M., & Pinter, N. (2012). *Hydraulic and flood-loss modeling of levee, floodplain, and river management strategies, Middle Mississippi River, USA*. 551–575. https://doi.org/10.1007/s11069-011-9938-x